PAL REPORT

COMPARSION OF VARIOUS PARAMETERS of 12-bit RIPPLE CARRY ADDER

Parameters	4-bit approximation	5-bit approximation	6-bit approximation	7-bit approximation	8-bit approximation
MEAN ERROR DISTANCE (MED)	3.5	7.5	15.5	31.5	63.5
NORMALISED MEAN ERROR DISTANCE (NMED)	0.00042798	0.000915639	0.001892323	0.003845684	0.007752411
ERROR RATE (total no. of cases =16777216)	6029312	7602176	8945664	10092544	11071488
ACCURACY	64.0625%	54.6875%	46.679%	39.84375%	34.0087%

PSNR COMPARISON BETWEEN THE IMAGES WITH THREE DIFFERENT FILTERS

PSNR comparison	Using the exact adder	Using the 4-bit approximate adder		Using the 5-bit approximate adder		Using the 6-bit approximate adder		Using the 7-bit approximate adder		Using the 8-bit approximate adder	
	Original i/p image v/s o/p of the kernel	Original i/p image v/s approx o/p image of the kernel	o/p image of exact adder v/s approx o/p image of the kernel	Original i/p image v/s approx o/p image of the kernel	o/p image of exact adder v/s approx o/p image of the kernel	Original i/p image v/s approx o/p image of the kernel	o/p image of exact adder v/s approx o/p image of the kernel	Original i/p image v/s approx o/p image of the kernel	o/p image of exact adder v/s approx o/p image of the kernel	Original i/p image v/s approx o/p image of the kernel	o/p image of exact adder v/s approx o/p image of the kernel
Box kernel	29.10 db	28.88 db	35.41 db	27.16 db	30.34 db	23.23 db	24.25 db	17.32 db	17.53 db	13.24 db	13.33 db
Laplacian kernel	6.44 db	6.31 db	13.42 db	6.46 db	13.33 db	6.87 db	12.89 db	6.98 db	11.38 db	6.73 db	10.00 db
Sobel kernel	6.52 db	6.58 db	23.40 db	6.76 db	21.43 db	7.06 db	18.11 db	7.04 db	13.67 db	7.07 db	11.64 db

SYNTHESIS RESULT OF 12 BIT ADDER USING 45 NM CMOS TECHNOLOGY USING CADENCE

	Cell count	Area (u m^2)	Leakage power (nW)	Internal power (nW)	Net power (nW)	Switching power (nW)	Critical path delay
Without approx	79	132.010	9.385	2894.658	868.248	3762.906	800 ps
4- bit approx	67	115.600	8.261	2345.973	712.149	3058.122	584 ps
5-bit approx	63	110.120	7.886	2161.933	659.807	2821.740	513 ps
6-bit approx	59	104.650	7.511	1977.148	607.259	2584.407	441 ps
7-bit approx	55	99.180	7.135	1792.098	554.625	2346.723	369 ps
8-bit approx	51	93.710	6.758	1608.861	502.440	2111.302	298 ps